

3 Textiles used in clothing can be made from:

- plants, for example cotton
- animals, for example wool and leather
- crude oil, for example nylon and polyester.

The photographs show cotton being grown and harvested.



(Source: © Design Pics Inc/Alamy Stock Photo)



(Source: © Jim Parkin/Alamy Stock Photo)

In 2022, it was estimated that 92 million tonnes of textile waste was produced in the world.

Textile waste is either burned or buried underground in landfill sites.

- (a) It is estimated that by 2030, there will be 134 million tonnes of textile waste produced each year.

Calculate this increase in textile waste.

Give your answer in standard form.

(1)

Answer tonnes

- (b) The production of textiles and the removal of textile waste is an example of human activity affecting the environment.

Which term is used to describe the effect of **human** activity on the environment?

(1)

- ☐ **A** antigenic
- ☐ **B** anthropogenic
- ☐ **C** epigenetic
- ☐ **D** polygenic

- (c) There are companies in Scandinavia that are producing sustainable textiles for clothing.

One company is recycling textile waste. It is removing dyes and making new fibres that can be woven.

Another company is using waste from the wood industry, wheat straw and leather scraps to make fibres.

- (i) Give the meaning of the term **sustainable**, as used in this context.

(1)

- * (ii) Discuss the extent to which these companies are reducing the conflict between human needs, climate change and conservation.

Use the information in the question, the photographs and your own knowledge to support your answer.

(6)

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(Total for Question 3 = 9 marks)